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SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – CONCEPT MAPPING

Course Code:	CSA1220	Course Title:	Computer Architecture
CO Assessed:	CO1 – Precision (BL4)	Assessment:	Concept Mapping
Weightage:	35%	Total Marks:	100
CO1 Statement:	Analyse the functional units of a computer system including CPU, memory, bus structures, and I/O components by examining instruction execution cycles, addressing modes, and performance metrics such as CPI, clock cycles, and benchmarking		
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Section / Batch:	2024	Date:	17-07-26

Instructions to Students

- Identify the key computer architecture concepts, components, and performance parameters mentioned in the given topic or scenario.
- Organize the identified concepts logically by establishing relationships between CPU, memory, I/O devices, bus structures, instruction execution cycle, and performance metrics.
- Represent the concepts using an appropriate concept map with clear nodes, connecting lines, and meaningful linking phrases.
- Demonstrate the hierarchy and interdependence among concepts, showing how one concept influences or interacts with another.
- Ensure the concept map is complete, accurate, well-structured, and easy to understand by using appropriate terminology and logical connections.

QUESTIONS

1. A software company is upgrading its office computers and must decide whether to use ARM Cortex, Intel Core, or AMD Ryzen processors for different workloads. Construct a concept map linking these modern architectures with their instruction execution, cache usage, processing capability, and application areas. Show how their architectural features influence performance and energy efficiency. How do modern processor architectures differ in meeting the requirements of mobile, desktop, and high-performance computing systems?
2. A computer technician is troubleshooting a system that stores machine instructions in memory but displays memory addresses in a debugger. Construct a concept map linking Binary and Hexadecimal Number Systems with instruction representation, memory storage, CPU interpretation, and data conversion. Show how instructions are represented, stored, and interpreted during execution. How do binary and hexadecimal number systems simplify memory organization, instruction representation, and debugging?
3. A manufacturing company is upgrading its automated inspection system and notices delays when multiple sensors and storage devices communicate with the processor simultaneously. Construct a concept map linking CPU, Memory, I/O Devices, Single-bus, Multi-bus, and Hierarchical Bus structures, showing how data and control signals are transferred. Illustrate the effect of each bus organization on communication efficiency. How do different bus structures influence system performance and reduce communication bottlenecks?
4. A programmer is developing an industrial automation application and must choose the most suitable instruction format and addressing mode for faster execution and efficient memory access. Construct a concept map linking Instruction Formats, Immediate, Direct, Register, Indirect, and Indexed Addressing Modes with Data Transfer, Arithmetic, and Control Instructions. Show how each addressing mode affects execution speed and memory access. How do instruction formats and addressing modes influence program execution and processor efficiency?
5. An embedded systems engineer is comparing the 8085 and 8086 microprocessors for a factory monitoring system that requires efficient instruction execution and memory organization. Construct a concept map linking their instruction sets, register organization, addressing modes, and memory organization, including memory segmentation in 8086. Show how these features affect program execution and system performance. How do the architectural differences between the 8085 and 8086 influence execution efficiency and memory management?

Code of Conduct

I ASWIN M (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ASWIN M

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Marks Evaluation:

Q. No	Concept Identification (25)	Linkages (25)	Organization (20)	Integration of Literature (20)	Clarity (10)	Total (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 25	/25	/ 20	/ 20	/ 10	/ 100

Course Coordinator

Faculty Incharge

15/07/2016

ASWIN M

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CSA1220

CO1 ASSESSMENT TOOL - 2

computer

Architecture

Modern Processor Architectures



→ Arm cortex is optimized for low power consumption, Intel core balanced desktop performance, AMD Ryzen offers multicore for gaming & HPC

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Binary & Hexadecimal Number Systems

Binary & hexadecimal number systems

Binary (0,1)
Base 2

Hexadecimal
Base 16 (0-9, A-F)

Instruction Representation

Memory Address Representation

Memory storage in words or bits

CPU Interpretation
Fetch → Decode → execute.

Data conversion
Binary ⇌ Hexadecimal

Advantages

- Easy memory
- compact
- Faster debugging.

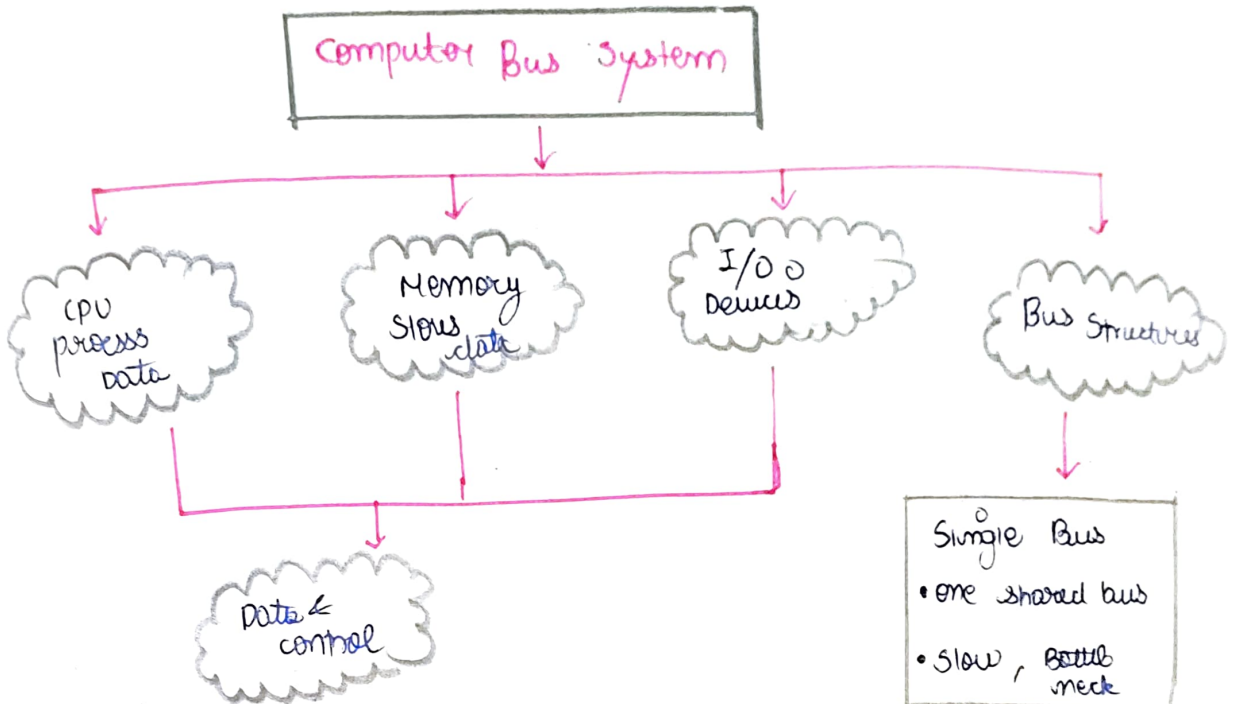
Binary

→ CPU's native language for storing and executing instructions

Hexadecimal

It provides a compact representation of binary making memory addresses and debugging easier.

Computer Bus System



Single Bus

- one shared bus
- slow, ~~bottle~~ neck

Multibus

Separate buses

Faster transfer

Hierarchical bus

Local + System

Highest efficiency

Single bus

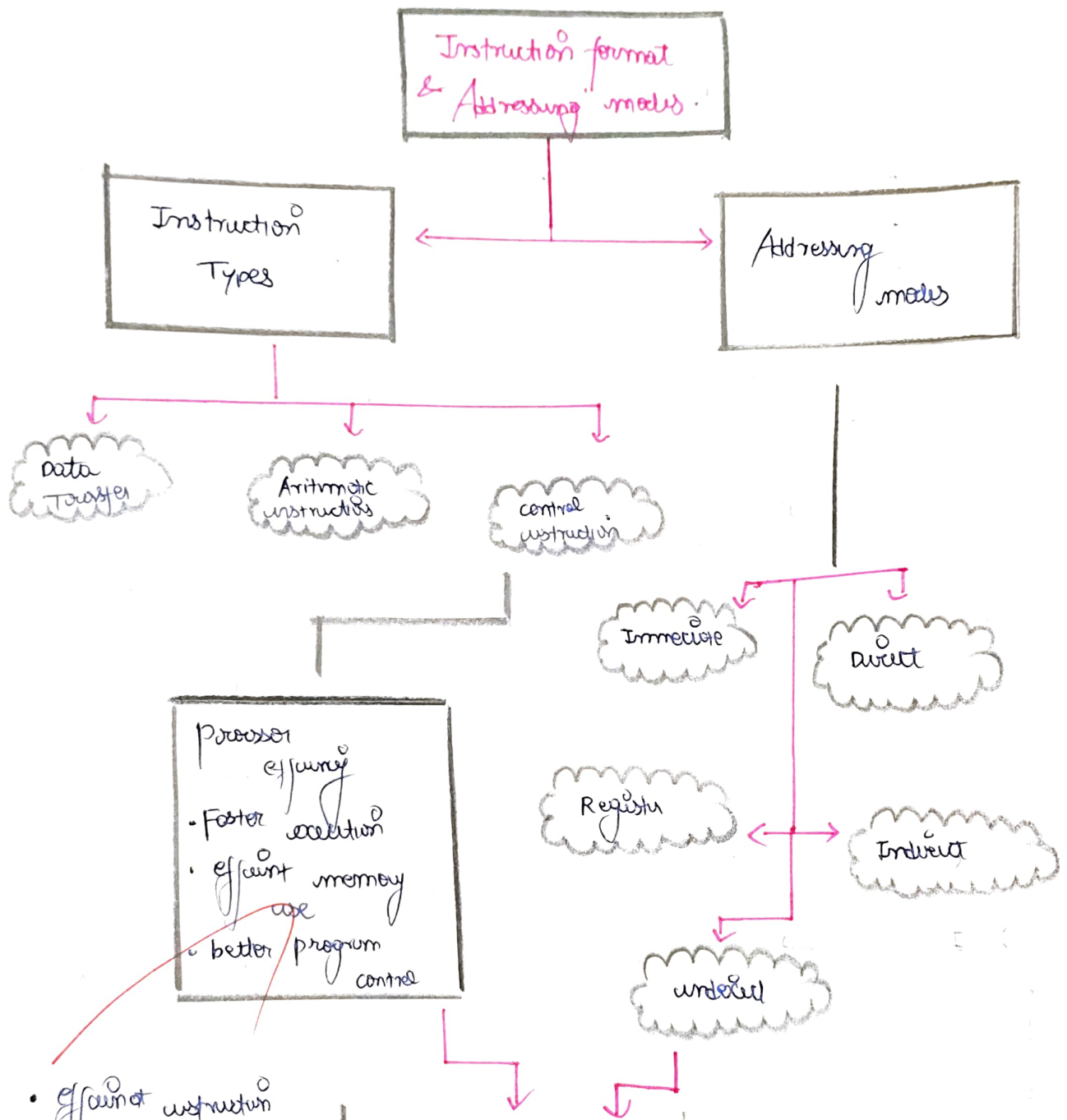
→ It causes data congestion whereas multibus and hierarchical bus allow simultaneous data transfers

→ This unidirectional communication speed and reduces bottle neck, increasing overall system performance.

System performance

- Single → low speed
- Multi → Better speed
- Hierarchical → Reduced Bottle necks & highest communication efficiency.

Instruction format & Addressing modes



Processor efficiency

- Faster execution
- efficient memory use
- better program control

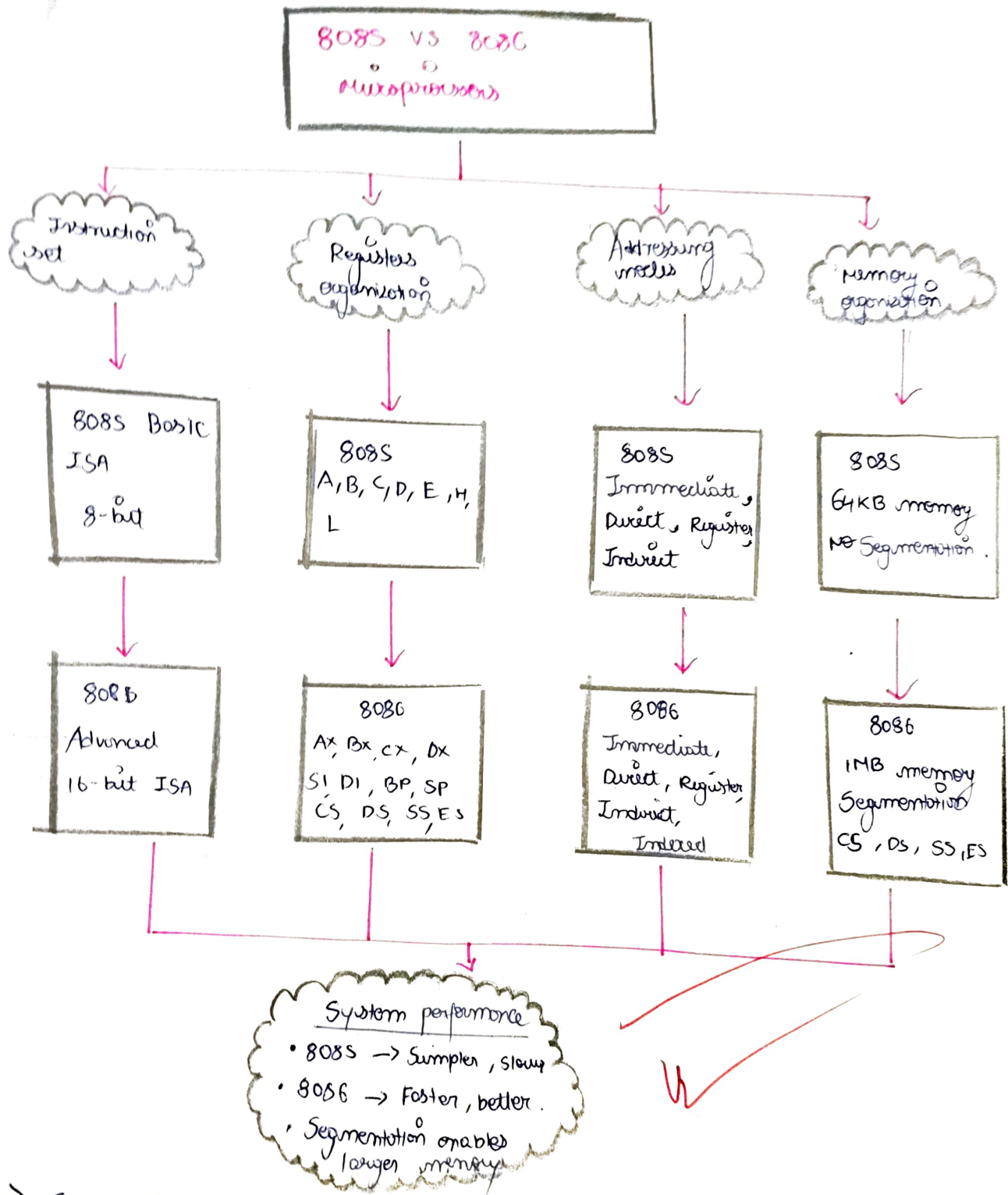
• efficient instruction formats and addressing modes improve program

• Register and immediate addressing are faster while indexed addressing is efficient for arrays.

Execution speed & Memory Access

- Immediate → Fast
- Register → Fastest
- Direct → Moderate
- Indirect → Slowest
- Indexed → Flexible

8085 VS 8086 Microprocessors



→ 8085 is a 8-bit processor with 64KB memory, making it suitable for simpler operations.

→ 8086 executes programs faster with 16-bit architecture and supports 1MB segmented memory.